Observational Data and Nonresponse

STA304 — Summer 2026

Learning Activity 9

Solutions

Part A

a) Not fully appropriate for a causal conclusion. Observational data can show whether study hours and grades are associated, but cannot by itself prove that studying causes higher grades.

b) Appropriate. This is an associational research question, so observational data can be used to investigate whether distance from campus is related to lecture attendance.

c) Not fully appropriate for a causal conclusion. Observational data can compare students who use AI tools and those who do not, but differences may be explained by other factors such as motivation, prior knowledge, or study habits.

d) Appropriate. This asks whether work hours and feeling overwhelmed are associated. Observational data can be used to study this relationship, but not necessarily to prove causation.

Part B

a) The true population mean is

$$\bar{y}_U = \frac{600(14) + 400(8)}{1000} = \frac{8400 + 3200}{1000} = 11.6.$$

b) The simple respondent average is

$$\bar{y}_R = 14.$$

c) The bias is

$$\text{Bias}(\bar{y}_R) = \bar{y}_R - \bar{y}_U = 14 - 11.6 = 2.4.$$

d) The respondent average is biased upward because respondents study more hours on average than nonrespondents. Since the nonrespondents are excluded, the survey over-represents students with higher study hours.

Part C

A response propensity is the probability that a sampled unit responds to the survey/gives data. In other words, it describes how likely someone is to provide a response, possibly depending on their characteristics.

Part D

a) MCAR. The missingness is caused by a random software glitch and does not depend on the respondent's characteristics or answers.

b) MAR, assuming that once age is accounted for, the probability of responding no longer depends on the student's exercise level. Since age is recorded for all sampled students, differences in response rates by age can be adjusted for. However, if students with different exercise levels are still more or less likely to respond even after conditioning on age, then the missingness would be NMAR.

c) NMAR. Missingness depends on the income value itself, which is the variable being asked about and is missing for some people.

d) NMAR. Missingness depends on course satisfaction, which is the main outcome of interest and is not observed for students who do not respond.

e) MAR, assuming that once sex and age group are accounted for, the probability of responding does not further depend on the unobserved survey responses. Since sex and age group are observed for both respondents and nonrespondents, differences in response rates across these groups can be adjusted for. However, if individuals with different values of the survey outcome remain more or less likely to respond even after conditioning on sex and age group, then the missingness would be NMAR.